derived quantities. Their corresponding units are called derived units.

- The area of a figure is the region covered by the boundary of the figure. Its SI unit is square metre or m^2.
- The area of irregularly shaped figures can be calculated with the help of a graph sheet.
- The amount of space occupied by a three dimensional object is known as its volume. The SI unit of volume is cubic metre or m^3.
- The volume of liquids are expressed in terms of litre. One litre = 1000 cc.
- The maximum volume of a liquid that a container can is known as the capacity of the container.
- Density of a substance is defined as the mass of the substance contained in unit volume (1 m^3).
- SI unit of density is kg/m^3. The CGS unit of density is g/cm^3. $1g/cm^3 = 10^3 kg/m^3$.
- The materials with higher density are called denser materials and the materials with lower density are called rarer materials.
- If the density of a solid is higher than that of a liquid, it sinks in that liquid. If the density of a solid is lower than that of a liquid, it floats in that liquid.
- Density = Mass / Volume

  Mass = Density × Volume

  Volume = Mass / Density
- One astronomical unit is defined as the average distance between the Earth and the Sun. 1 AU = 149.6×10^6 km = 1.496×10^{11} m.
- One light year is defined as the distance travelled by light in vacuum during the period of one year. 1 Light year = 9.46×10^{15} m.

Evaluation

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I. Choose the best answer.

1. Which of the following is a derived quantity?
   a) mass b) time
   c) area d) length
2. Which of the following is correct?
   a) 1L = 1cc b) 1L = 10 cc
   c) 1L = 100 cc d) 1L = 1000 cc
3. SI unit of density is
   a) kg/m^2 b) kg/m^3 c) kg/m d) g/m^3
4. Two spheres have mass and volume in the ratio 2:1. The ratio of their density is
   a) 1:2 b) 2:1 c) 4:1 d) 1:4
5. Light year is the unit of
   a) distance b) time
   c) density d) Both length and time

II. Fill in the blanks.

1. Volume of irregularly shaped objects are measured using the law of _____.
2. One cubic metre is equal to _____ cubic centimetre.
3. Density of mercury is _____.
4. One astronomical unit is equal to _____.
5. The area of a leaf can be measured using a _____.

III. State true or false. If false, correct the statement.

1. The region covered by the boundary of a plane figure is called its volume.

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